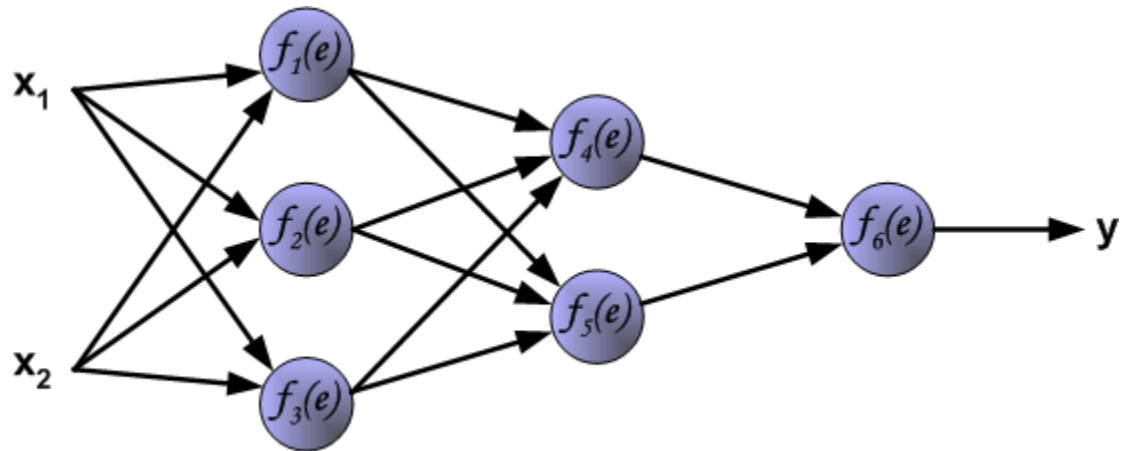
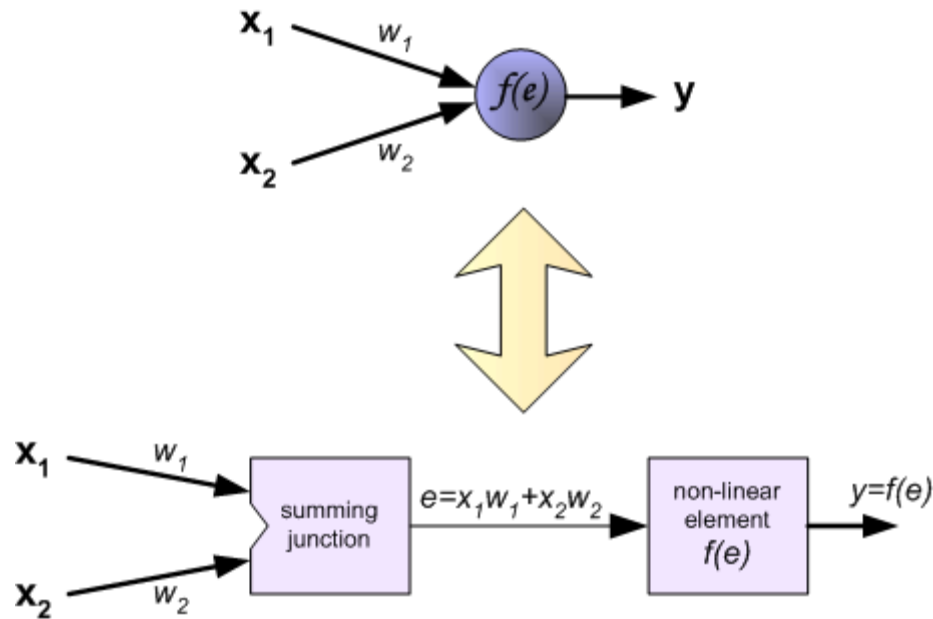


Yapay Sinir Ağları
Artificial Neural Networks (ANN)
Backpropagation Algoritması

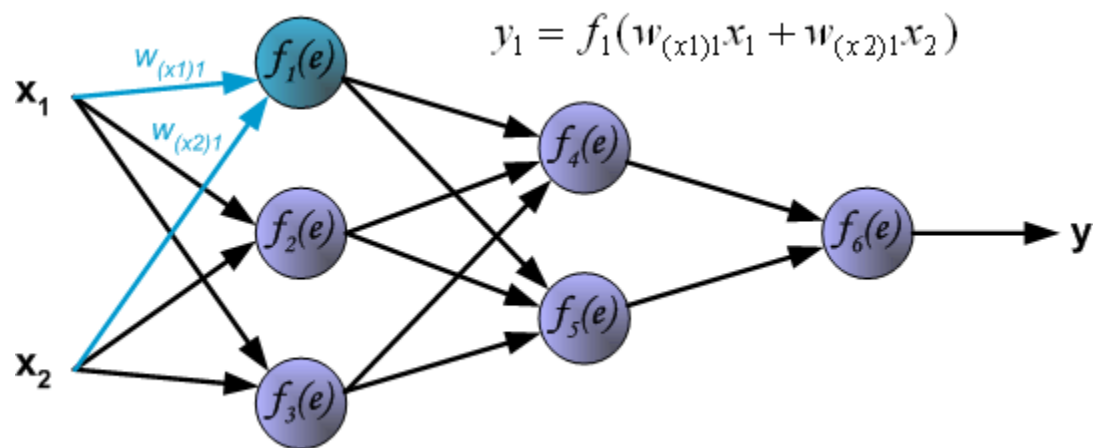
Backpropagation Adımları



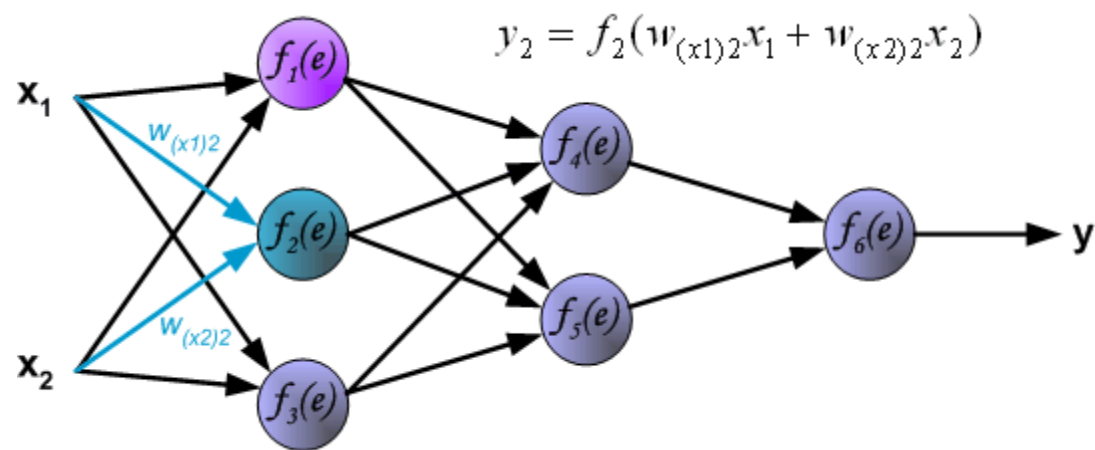
Backpropagation Adımları



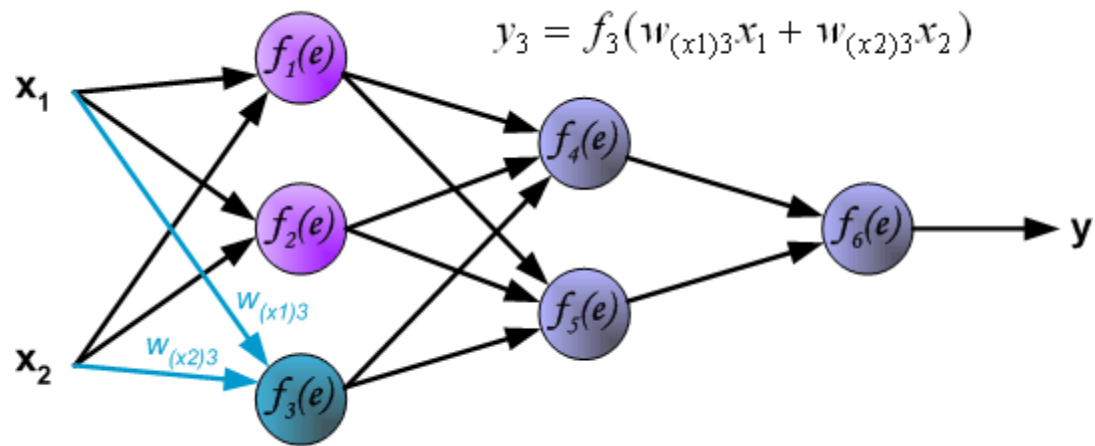
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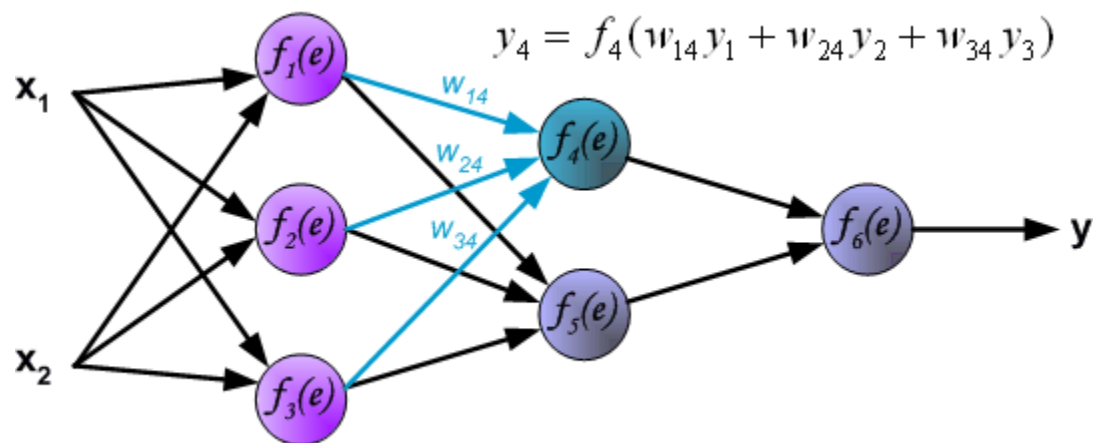
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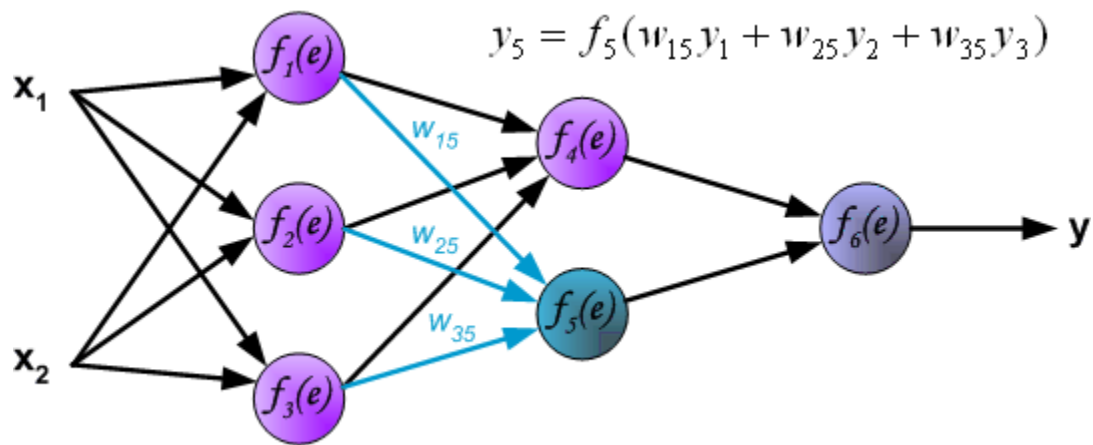
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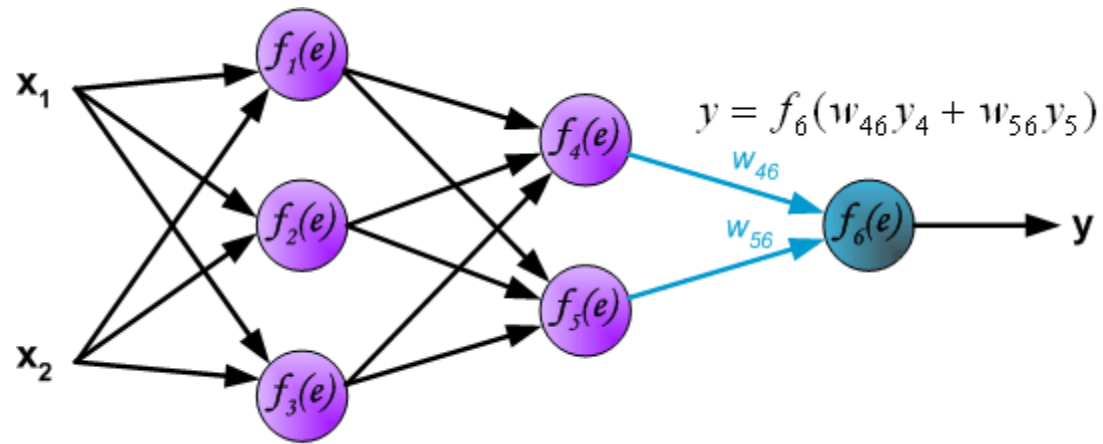
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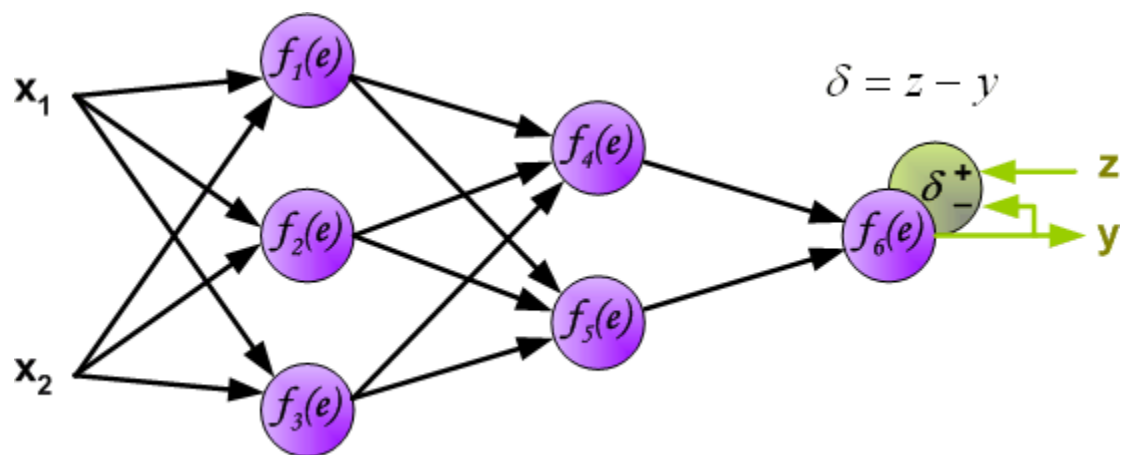
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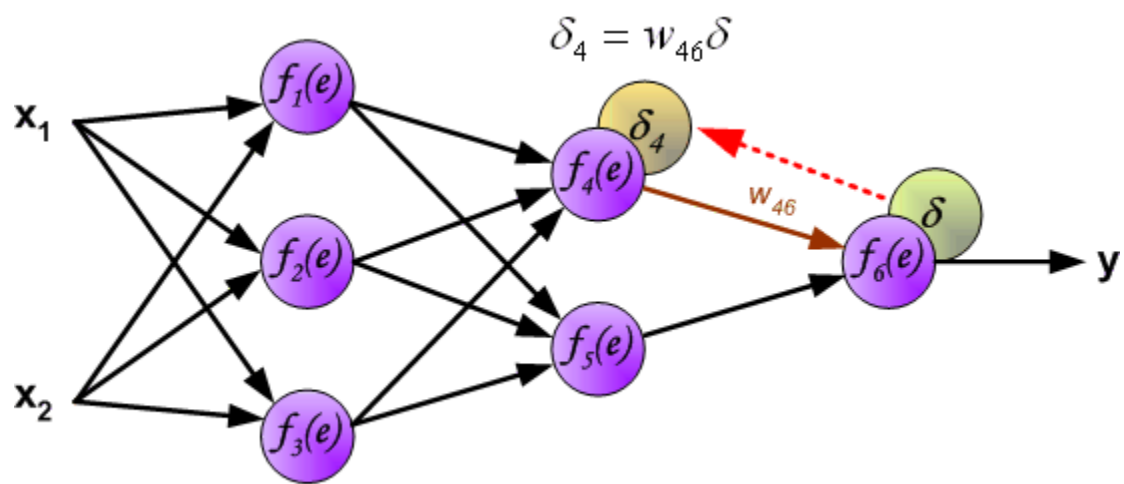
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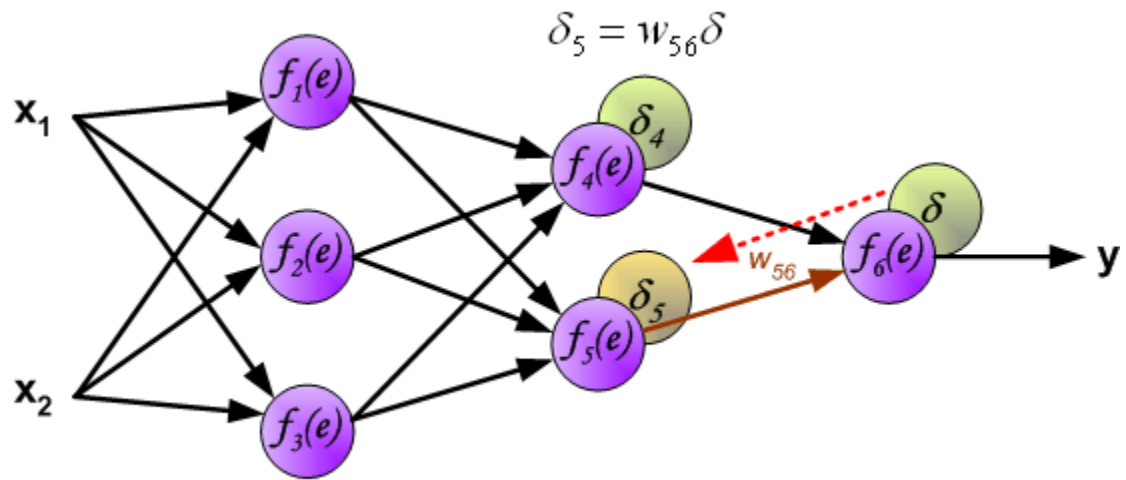
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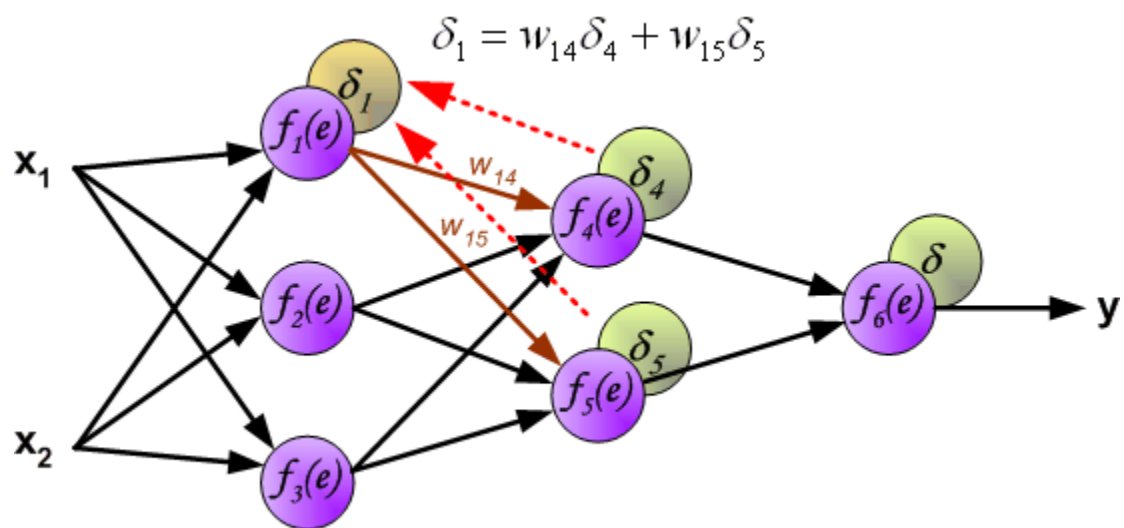
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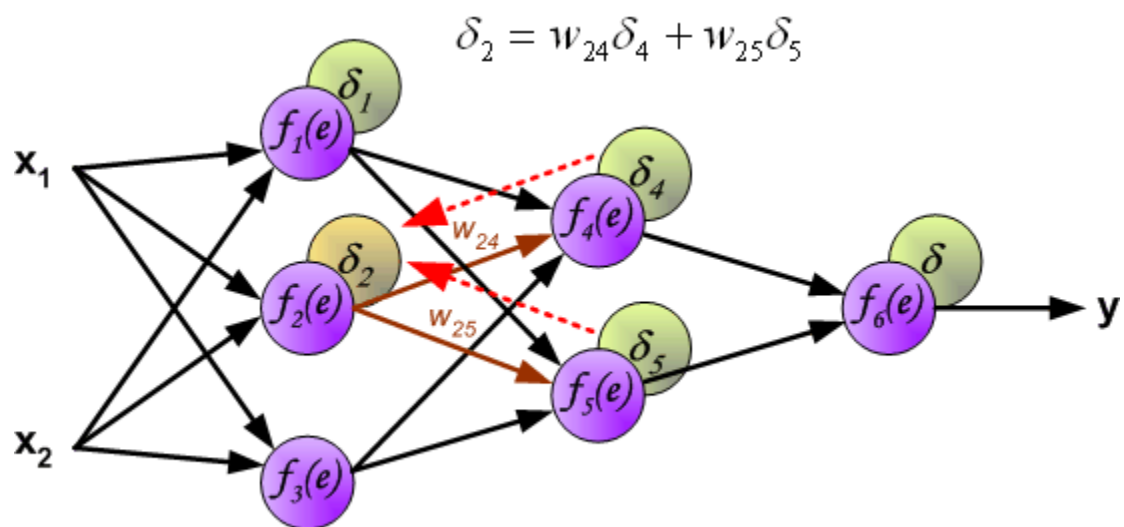
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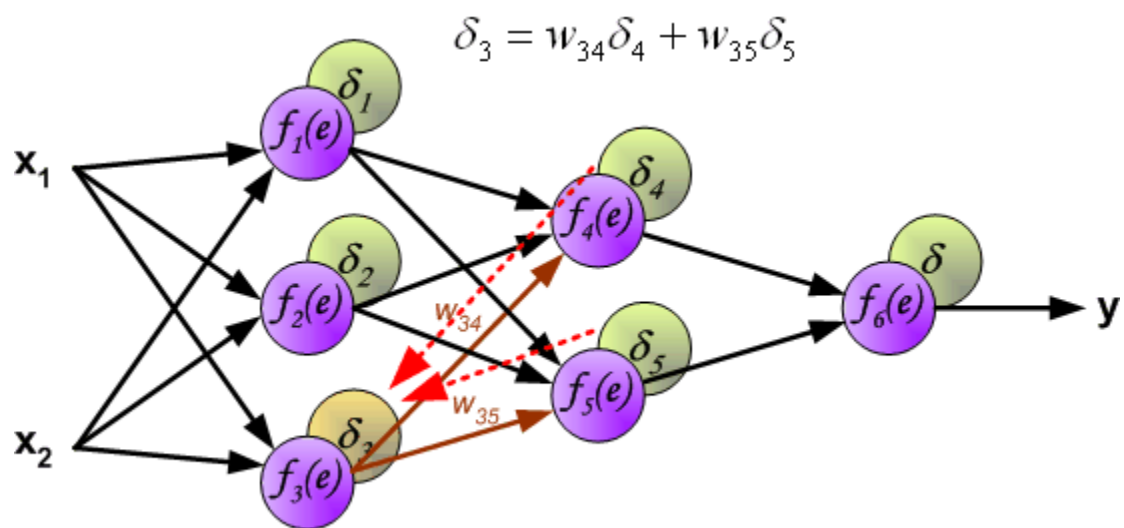
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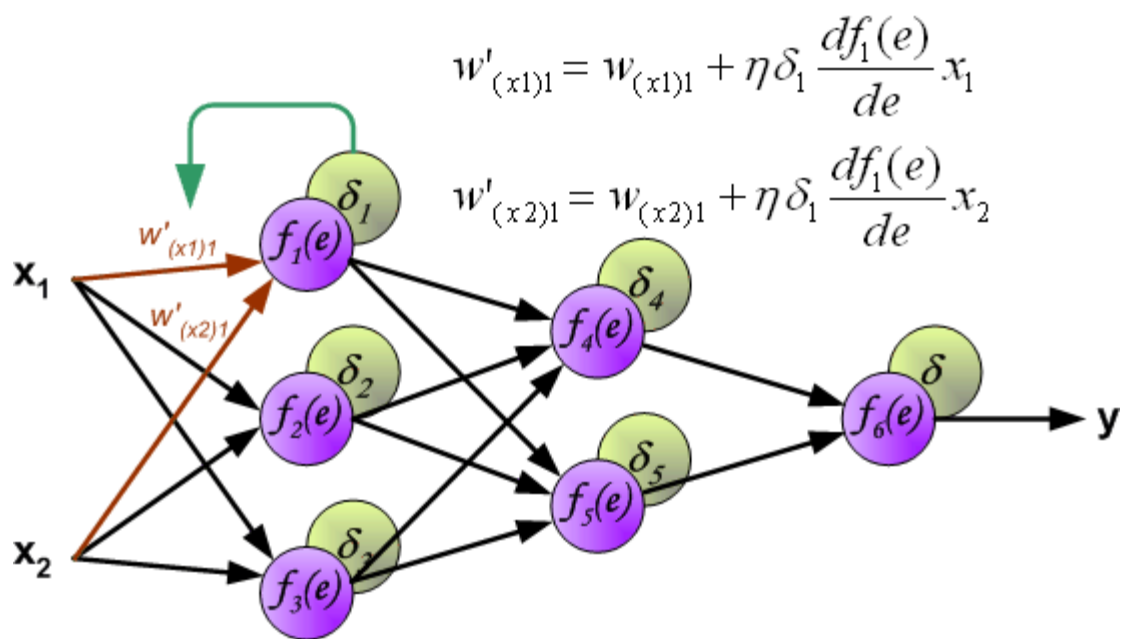
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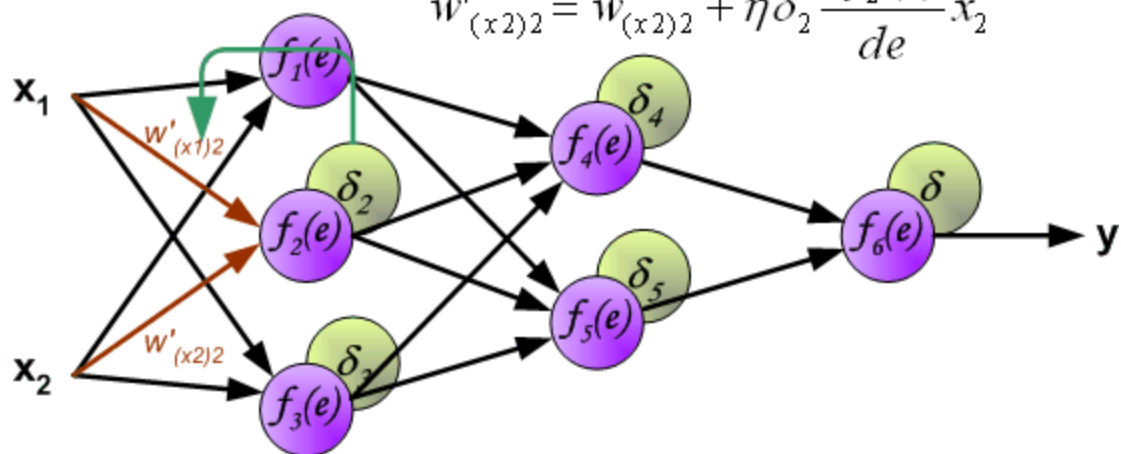
...



...

$$w'_{(x1)2} = w_{(x1)2} + \eta \delta_2 \frac{df_2(e)}{de} x_1$$

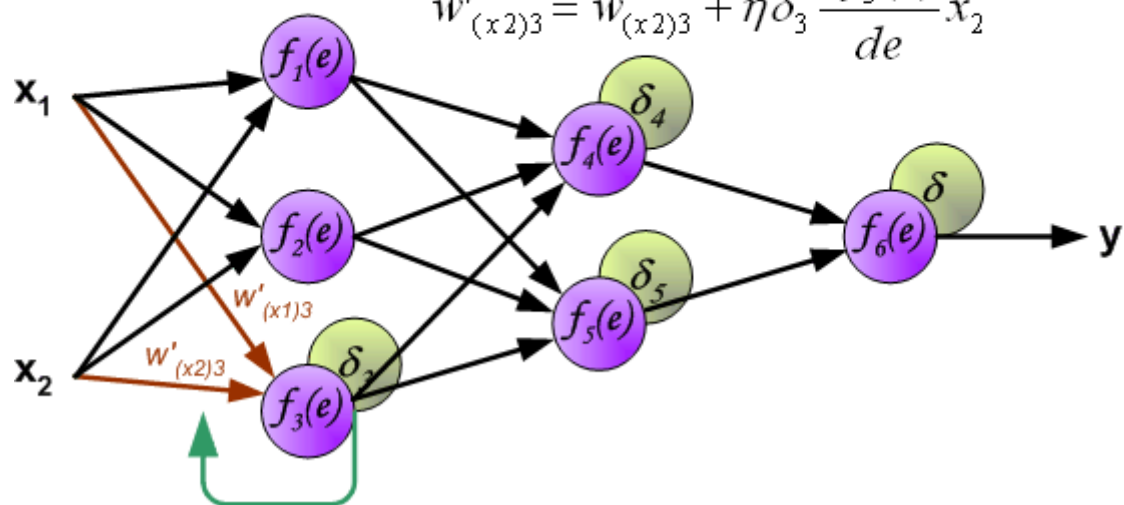
$$w'_{(x2)2} = w_{(x2)2} + \eta \delta_2 \frac{df_2(e)}{de} x_2$$



...

$$w'_{(x1)3} = w_{(x1)3} + \eta \delta_3 \frac{df_3(e)}{de} x_1$$

$$w'_{(x2)3} = w_{(x2)3} + \eta \delta_3 \frac{df_3(e)}{de} x_2$$

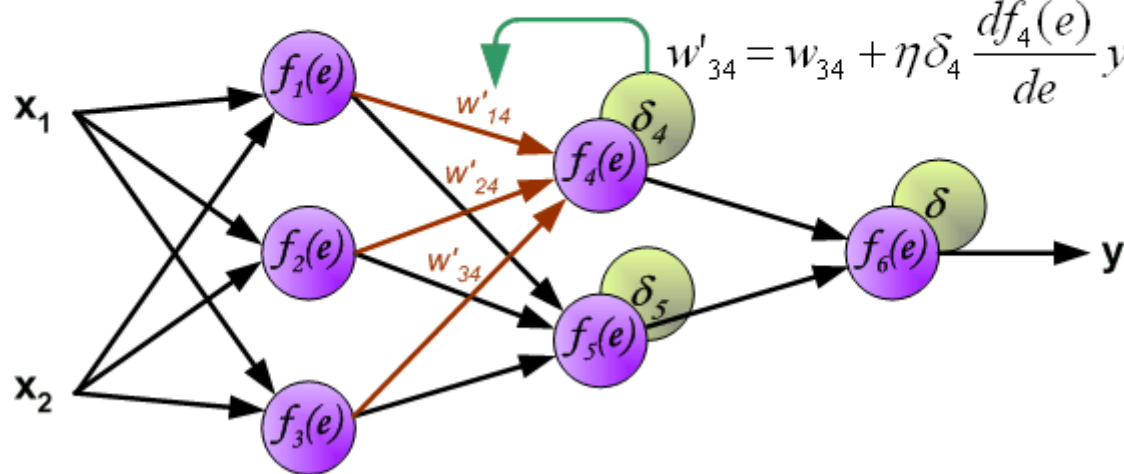


...

$$w'_{14} = w_{14} + \eta \delta_4 \frac{df_4(e)}{de} y_1$$

$$w'_{24} = w_{24} + \eta \delta_4 \frac{df_4(e)}{de} y_2$$

$$w'_{34} = w_{34} + \eta \delta_4 \frac{df_4(e)}{de} y_3$$

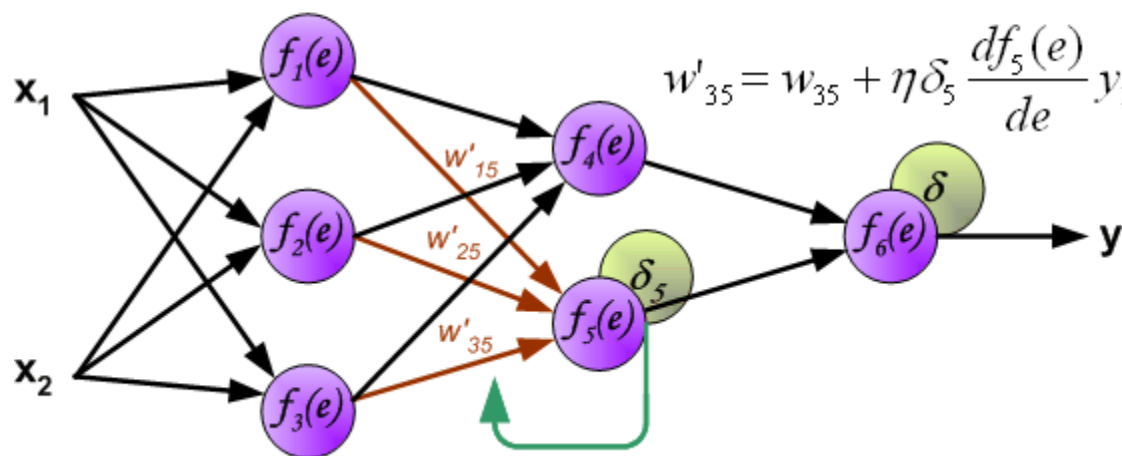


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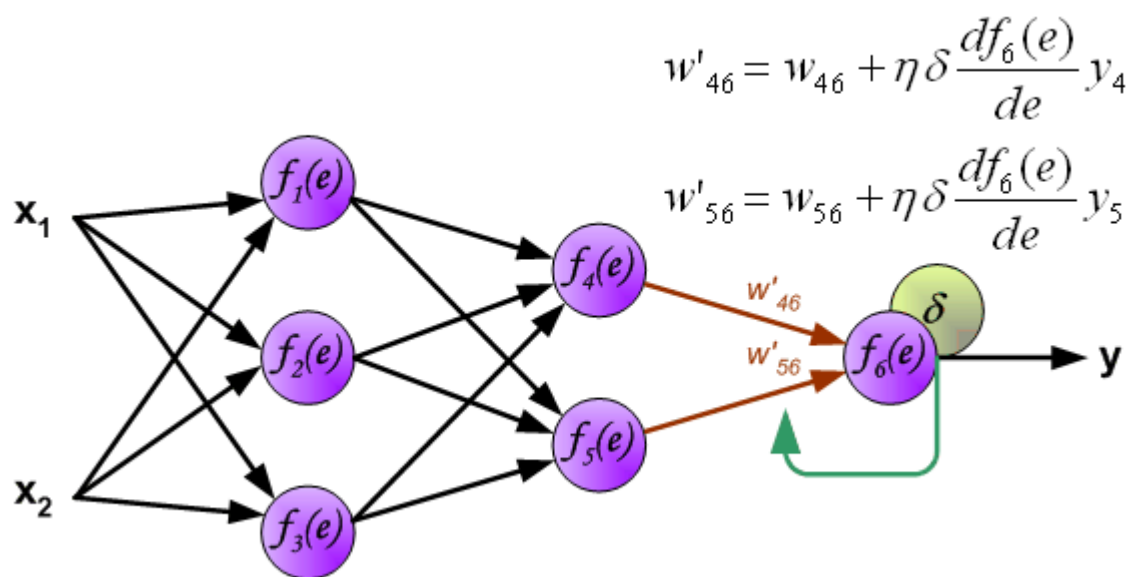
$$w'_{15} = w_{15} + \eta \delta_5 \frac{df_5(e)}{de} y_1$$

$$w'_{25} = w_{25} + \eta \delta_5 \frac{df_5(e)}{de} y_2$$

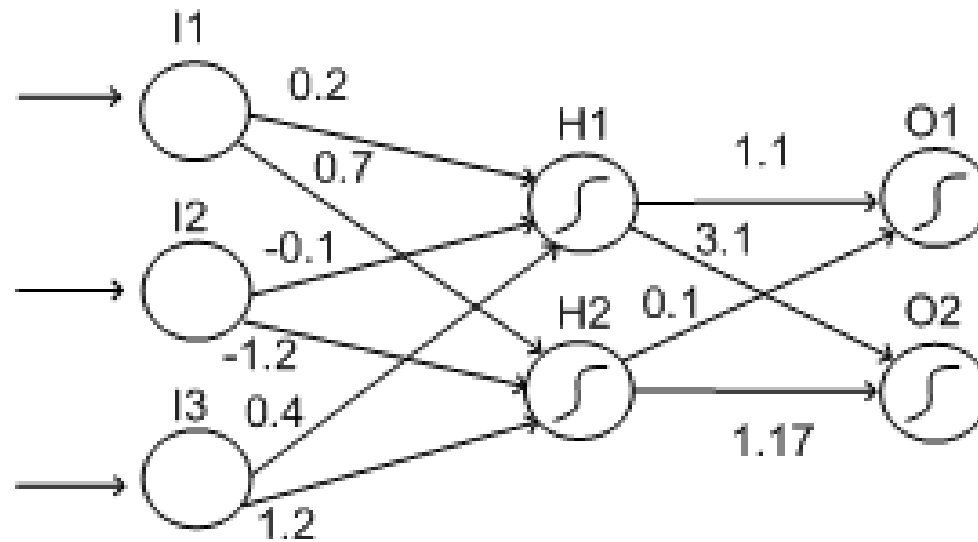
$$w'_{35} = w_{35} + \eta \delta_5 \frac{df_5(e)}{de} y_3$$



...



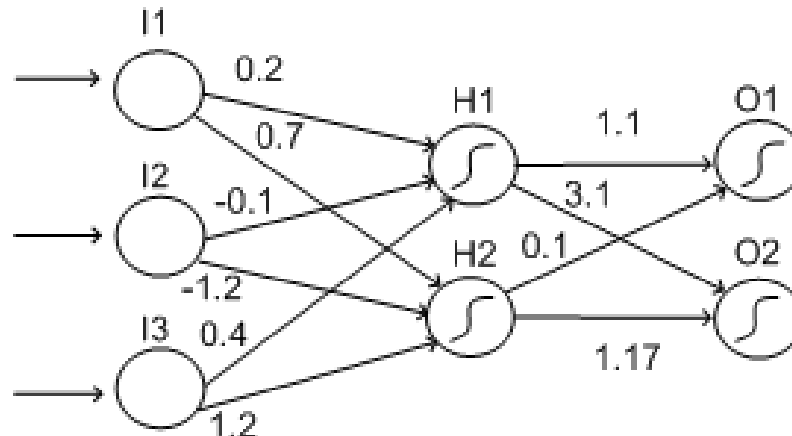
MLP Örnek



Örnekleri ileri sürmek

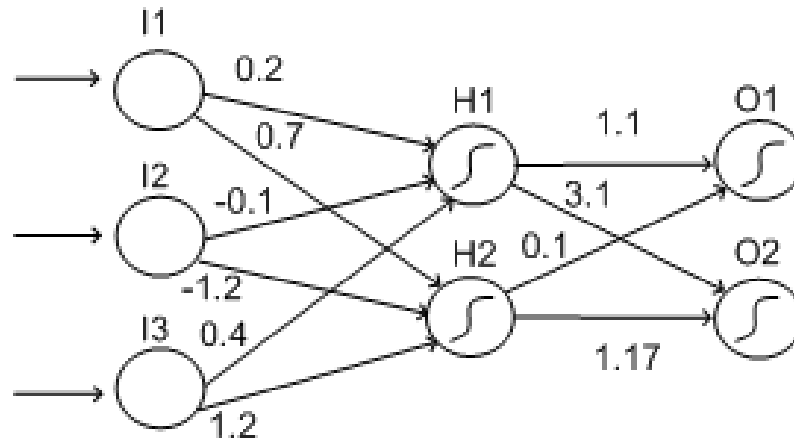
Ağa sunulan örnekler: E(10,30,20)

- Gizli katmanlarda ağırlıklı toplamlar:
 - $S_{H1} = (0.2*10) + (-0.1*30) + (0.4*20) = 2-3+8 = 7$
 - $S_{H2} = (0.7*10) + (-1.2*30) + (1.2*20) = 7-6+24 = -5$
- Bir sonraki adımda gizli katman çıkışları hesaplanır:
 - $\sigma(S) = 1/(1 + e^{-S})$
 - $\sigma(S_{H1}) = 1/(1 + e^{-7}) = 1/(1+0.000912) = 0.999$
 - $\sigma(S_{H2}) = 1/(1 + e^5) = 1/(1+148.4) = 0.0067$



...

- Ağırlıklı toplamaları çıkış katmanında hesapla:
 - $S_{O1} = (1.1 * 0.999) + (0.1 * 0.0067) = 1.0996$
 - $S_{O2} = (3.1 * 0.999) + (1.17 * 0.0067) = 3.1047$
- Son olarak ANN çıkışını hesapla:
 - $\sigma(S_{O1}) = 1/(1+e^{-1.0996}) = 1/(1+0.333) = 0.750$
 - $\sigma(S_{O2}) = 1/(1+e^{-3.1047}) = 1/(1+0.045) = 0.957$

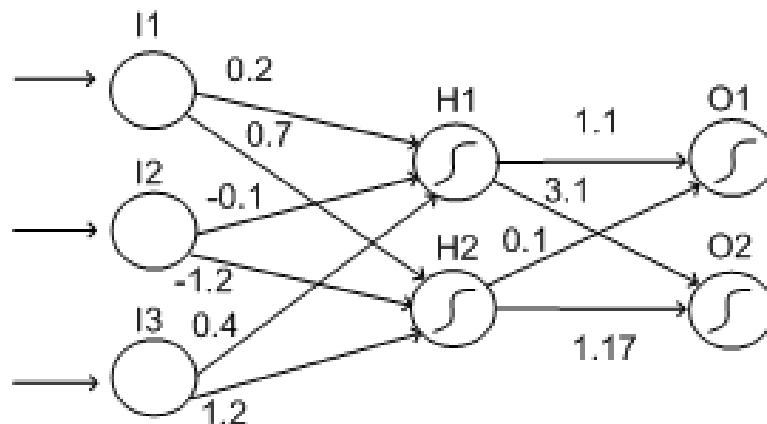


Örnekleri ileri doğru sürdüğümüzde elde edilen hesaplamalar

Ağa sunulan örnekler E(10,30,20)

$$\eta = 0.1$$

Input units		Hidden units			Output units		
Unit	Output	Unit	Weighted Sum Input	Output	Unit	Weighted Sum Input	Output
I1	10	H1	7	0.999	O1	1.0996	0.750
I2	30	H2	-5	0.0067	O2	3.1047	0.957
I3	20						



Çıkış Birimlerinin Hata Değerleri

- $o_1(E)=0.750$, $o_2(E)=0.957$

Ağdaki her k çıkış birimi için, bu çıkış biriminin δ_k hata terimini hesapla.

$$\circ \quad \delta_k = o_k(1 - o_k)(t_k - o_k)$$

$$\delta_{O1} = o_1(E)(1 - o_1(E))(t_1(E) - o_1(E)) = 0.750(1-0.750)(1-0.750) = 0.0469$$

$$\delta_{O2} = o_2(E)(1 - o_2(E))(t_2(E) - o_2(E)) = 0.957(1-0.957)(0-0.957) = -0.0394$$

Gizli Birimler için Hata Değerleri

ağdaki her h gizli katmanı için, gizli birim δ_k hatasını hesapla

$$\circ \quad \delta_h = o_h(1 - o_h) \sum_{k \in \text{outputs}} w_{kh} \delta_k$$

- $\delta_{O1} = 0.0469$ ve $\delta_{O2} = -0.0394$
- $h_1(E) = 0.999$ ve $h_2(E) = 0.0067$
- H1 için:
 - $(w_{11} * \delta_{O1}) + (w_{12} * \delta_{O2}) = (1.1 * 0.0469) + (3.1 * -0.0394) = -0.0706$
 - Ve bu değeri, $h_1(E)(1-h_1(E))$ ile çarpalım, sonuç:
 - $-0.0706 * (0.999 * (1-0.999)) = 0.0000705 = \delta_{H1}$
- H2 için:
 - $(w_{21} * \delta_{O1}) + (w_{22} * \delta_{O2}) = (0.1 * 0.0469) + (1.17 * -0.0394) = -0.0414$
 - Ve bu değeri, $h_2(E)(1-h_2(E))$ ile çarpalım, sonuç:
 - $-0.0414 * (0.067 * (1-0.067)) = -0.00259 = \delta_{H2}$

Ağırlık güncellemeleri için hesaplamalar

- Giriş ve gizli katman arasındaki ağırlık değerleri:

Input unit	Hidden unit	η	δ_H	x_i	$\Delta = \eta * \delta_H * x_i$	Old weight	New weight
I1	H1	0.1	-0.0000705	10	-0.0000705	0.2	0.1999295
I1	H2	0.1	-0.00259	10	-0.00259	0.7	0.69741
I2	H1	0.1	-0.0000705	30	-0.0002115	-0.1	-0.1002115
I2	H2	0.1	-0.00259	30	-0.00777	-1.2	-1.20777
I3	H1	0.1	-0.0000705	20	-0.000141	0.4	0.39999
I3	H2	0.1	-0.00259	20	-0.00518	1.2	1.1948

her ağ bağlantısı w_{ji} 'yi güncelle

- $w_{ji} = w_{ji} + \Delta w_{ji}$
- $\Delta w_{ji} = \eta \delta_j x_{ji}$

Ağırlık güncellemeleri için hesaplamalar

- Gizli ve çıkış katmanı arasındaki ağırlık değerleri:

Hidden unit	Output unit	η	δ_O	$h_i(E)$	$\Delta = \eta * \delta_O * h_i(E)$	Old weight	New weight
H1	O1	0.1	0.0469	0.999	0.000469	1.1	1.100469
H1	O2	0.1	-0.0394	0.999	-0.00394	3.1	3.0961
H2	O1	0.1	0.0469	0.0067	0.00314	0.1	0.10314
H2	O2	0.1	-0.0394	0.0067	-0.0000264	1.17	1.16998

her ağ bağlantısı w_{ji} 'yi güncelle

- $w_{ji} = w_{ji} + \Delta w_{ji}$

- $\Delta w_{ji} = \eta \delta_j x_{ji}$

References

- T.M. Mitchell, Machine Learning, McGraw Hill, 1997.
- E.Alpaydin, Introduction to Machine Learning, MIT Press, 2010.
- Daniel T. Larose, Discovering Knowledge in Data, Wiley, 2005.
- http://galaxy.agh.edu.pl/~vlsi/AI/backp_t_en/backprop.html